Course: Cryptography and Network Security
Code: CSN15403
Branch: B.Tech -5th Semester

Lecture – 6 : MATHEMATICS OF CRYPTOGRAPHY - PART II
ALGEBRAIC STRUCTURES

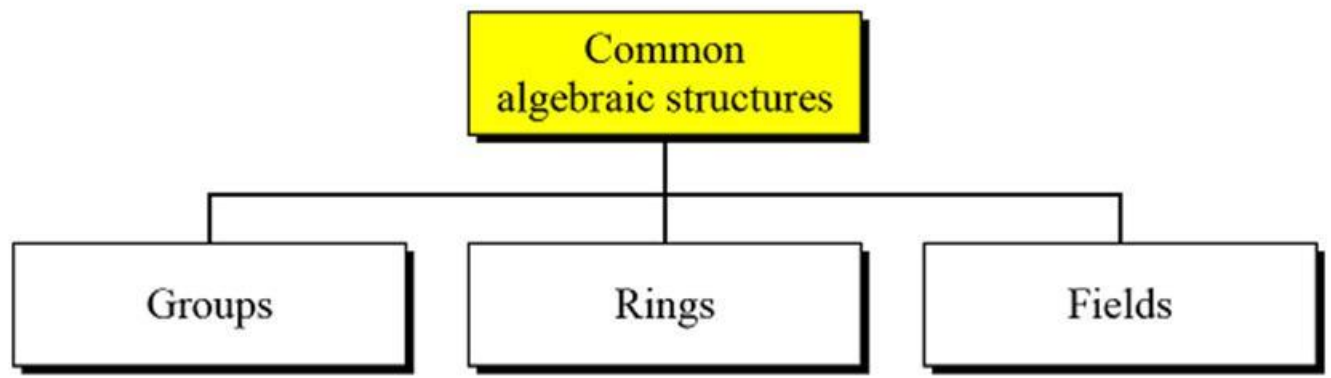
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ALGEBRAIC STRUCTURES

- Cryptography requires sets of integers and specific operations that are defined for those sets.
- The combination of the set and the operations that are applied to the elements of the set is called an algebraic structure.
- Three common algebraic structures:
  - Groups
  - Rings, and
  - Fields.



Groups

- A group (G) is a set of elements with a binary operation (•) that satisfies four properties (or axioms).
- Closure
  - If a and b are elements of G, then c = a•b is also an element of G.
- Associativity
  - If a, b and c are elements of G, then (a•b) •c=a•(b•c)
- Existence of identity
  - For all a in G, there exist an element e, called the identity element, such that e•a=a•e=a
- Existence of inverse
  - For each a in G, there exists an element a', called the inverse of a, such that a•a'=a'•a=e

Groups

- A Commutative group (Abelian group) is group in which the operator satisfies four properties plus an extra property that is commutativity.
  - For all a and b in G, we have a • b = b • a
- Application
  - Although a group involves a single operation, the properties imposed on the operation allow the use of a pair of operations!!!!
- Example
  - *The set of residue integers with the addition operator,*
    $G = < Z_n , + >$,
    is a commutative group.
    Check the properties….

Groups

Let us check the properties.

1. Closure is satisfied. The result of adding two integers in Zn is another integer in Z_n.

2. Associativity is satisfied. The result of 4 + (3 + 2) is the same as (4 + 3) + 2.

3. Commutativity is satisfied. We have 3 + 5 = 5 + 3.

4. The identify element is 0. We have 3 + 0 = 0 + 3 = 3.

5. Every element has an additive inverse. The inverse of an element is its complement. For example, the inverse of 3 is −3 (n − 3 in Zn) and the inverse of−3 is 3. The inverse allows us to perform subtraction on the set.

Groups

- The set Z_n^* with the multiplication operator, $G = \langle Z_n^*, \times \rangle$, is also an abelian group.
- We can perform multiplication and division on the elements of this set without moving out of the set.
- It is easy to check the first three properties.
- The identity element is 1.
- Each element has an inverse that can be found according to the extended Euclidean algorithm.

Groups

- Let us define a set G = < {a, b, c, d}, •> and the operation

•	*a*	*b*	*c*	*d*
a	*a*	*b*	*c*	*d*
b	*b*	*c*	*d*	*a*
c	*c*	*d*	*a*	*b*
d	*d*	*a*	*b*	*c*

Check for properties….

- Is the group abelian????

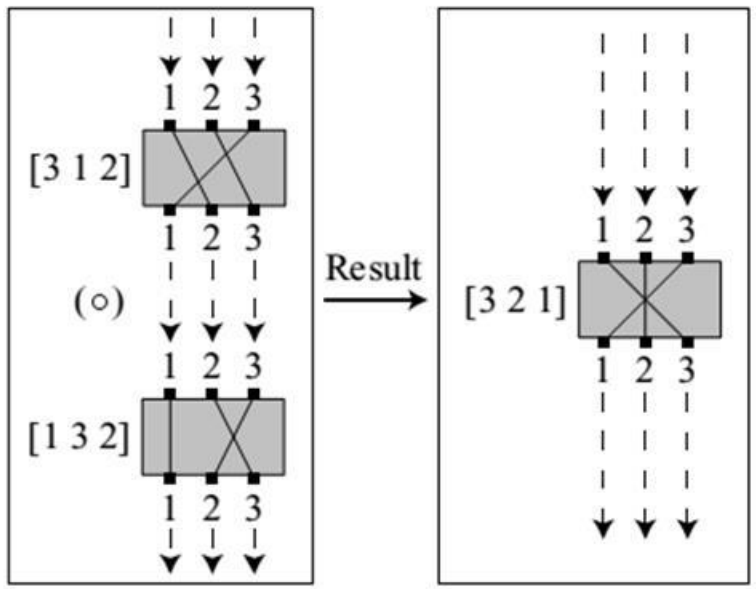
Groups

•	a	b	c	d
a	a	b	c	d
b	b	c	d	a
c	c	d	a	b
d	d	a	b	c

- This is an abelian group. All five properties are satisfied:
  1. Closure is satisfied. Applying the operation on any pair of elements result in another elements in the set.
  2. Associativity is also satisfied. To prove this we need to check the property for any combination of three elements. For example, (a + b) + c = a + (b + c) = d.
  3. The operation is commutative. We have a + b = b + a.
  4. The group has an identity element, which is a.
  5. Each element has an inverse. The inverse pairs can be found by finding the identity in each row (shaded). The pairs are (a, a), (b, d), (c, c).

Groups

- A very interesting group is the permutation group.
- The set is the set of all permutations, and the operation is composition: applying one permutation after another.



$$[3\ 2\ 1] = [3\ 1\ 2] \circ [1\ 3\ 2]$$

Read from right to left:
The first permutation is [1 3 2] followed by [3 1 2];
the result is [3 2 1].

Operation table for permutation group

°	[1 2 3]	[1 3 2]	[2 1 3]	[2 3 1]	[3 1 2]	[3 2 1]
[1 2 3]	[1 2 3]	[1 3 2]	[2 1 3]	[2 3 1]	[3 1 2]	[3 2 1]
[1 3 2]	[1 3 2]	[1 2 3]	[2 3 1]	[2 1 3]	[3 2 1]	[3 1 2]
[2 1 3]	[2 1 3]	[3 1 2]	[1 2 3]	[3 2 1]	[1 3 2]	[2 3 1]
[2 3 1]	[2 3 1]	[3 2 1]	[1 3 2]	[3 1 2]	[1 2 3]	[2 1 3]
[3 1 2]	[3 1 2]	[2 1 3]	[3 2 1]	[1 2 3]	[2 3 1]	[1 3 2]
[3 2 1]	[3 2 1]	[2 3 1]	[3 1 2]	[1 3 2]	[2 1 3]	[1 2 3]

Check for properties….

- Is the group abelian????

Permutation group

- Closure is satisfied.
- Associativity is also satisfied.
- The commutative property is not satisfied.
- The set has an identity element, which is [1 2 3] (no permutation).
- Each element has an inverse.
- Set of permutations with the composition operation is a group.
- This implies that using two permutations one after another cannot strengthen the security of a cipher, because we can always find a permutation that can do the same job because of the closure property.

Groups

- Finite Group
  - If the set has a finite number of elements; otherwise, it is an infinite group.
- Order of a Group |G|
  - The number of elements in the group.
  - If the group is finite, its order is finite
- Subgroups
  - A subset H of a group G is a subgroup of G if H itself is a group with respect to the operation on G
  - If G=<S, •> is a group, H=<T, •> is a group under the same operation, and T is a nonempty subset of S, then H is a subgroup of G
  - If a and b are members of both groups, then c=a•b is also member of both groups
  - The group share the same identity element
  - If a is a member of both groups, the inverse of a is also a member of both groups
  - The group made of the identity element of G, H=<{e}, •>, is a subgroup of G
  - Each group is a subgroup of itself

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Groups

- Is the group H = $<Z_{10}, +>$ a subgroup of the group G = $<Z_{12}, +>$?
- The answer is no. Although H is a subset of G, the operations defined for these two groups are different. The operation in H is addition modulo 10; the operation in G is addition modulo 12.

Groups

- Cyclic subgroups
  - If a subgroup of a group can be generated using the power of an element, the subgroup is called the cyclic subgroup.

$$a^n \rightarrow a \bullet a \bullet \ldots \bullet a \quad (n \text{ times})$$

Groups

- Four cyclic subgroups can be made from the group G = $\langle Z_6, + \rangle$.
- They are $H_1 = \langle \{0\}, + \rangle$, $H_2 = \langle \{0, 2, 4\}, + \rangle$, $H_3 = \langle \{0, 3\}, + \rangle$, and $H_4 = G$.

$0^0 \bmod 6 = 0$

$1^0 \bmod 6 = 0$
$1^1 \bmod 6 = 1$
$1^2 \bmod 6 = (1 + 1) \bmod 6 = 2$
$1^3 \bmod 6 = (1 + 1 + 1) \bmod 6 = 3$
$1^4 \bmod 6 = (1 + 1 + 1 + 1) \bmod 6 = 4$
$1^5 \bmod 6 = (1 + 1 + 1 + 1 + 1) \bmod 6 = 5$

$2^0 \bmod 6 = 0$
$2^1 \bmod 6 = 2$
$2^2 \bmod 6 = (2 + 2) \bmod 6 = 4$

$3^0 \bmod 6 = 0$
$3^1 \bmod 6 = 3$

$4^0 \bmod 6 = 0$
$4^1 \bmod 6 = 4$
$4^2 \bmod 6 = (4 + 4) \bmod 6 = 2$

$5^0 \bmod 6 = 0$
$5^1 \bmod 6 = 5$
$5^2 \bmod 6 = 4$
$5^3 \bmod 6 = 3$
$5^4 \bmod 6 = 2$
$5^5 \bmod 6 = 1$

Groups

- Exercise:
  – Find out the cyclic subgroups for group G = $<Z_{10}^*, \times>$.

Groups

- Three cyclic subgroups can be made from the group G = $\langle Z_{10}^*, \times \rangle$.
- G has only four elements: 1, 3, 7, and 9.
- The cyclic subgroups are $H_1 = \langle \{1\}, \times \rangle$, $H_2 = \langle \{1, 9\}, \times \rangle$, and $H_3 = G$.

$$1^0 \bmod 10 = 1$$

$$3^0 \bmod 10 = 1$$
$$3^1 \bmod 10 = 3$$
$$3^2 \bmod 10 = 9$$
$$3^3 \bmod 10 = 7$$

$$7^0 \bmod 10 = 1$$
$$7^1 \bmod 10 = 7$$
$$7^2 \bmod 10 = 9$$
$$7^3 \bmod 10 = 3$$

$$9^0 \bmod 10 = 1$$
$$9^1 \bmod 10 = 9$$

Groups

- Cyclic group
  – A cyclic group is a group that is its own cyclic subgroup.

$$\{e, g, g^2, \ldots, g^{n-1}\}, \text{ where } g^n = e$$

- Example:
  - Three cyclic subgroups can be made from the group G = $<Z_{10}^*, \times>$.
  - The cyclic subgroups are $H_1 = <\{1\}, \times>$, $H_2 = <\{1, 9\}, \times>$, and $H_3 = G$.
  - The group G = $<Z_{10}^*, \times>$ is a cyclic group with two generators, $g = 3$ and $g = 7$.
  - The group G = $<Z_6, +>$ is a cyclic group with two generators, $g = 1$ and $g = 5$.

Lagrange's Theorem

- Assume that G is a group, and H is a subgroup of G.
- If the order of G and H are |G| and |H|, respectively, then, based on this theorem, |H| divides |G| .
- Order of an Element
  – The order of an element is the order of the cyclic group it generates.
- Example:
  - In the group G = $<Z_6, +>$, the orders of the elements are:
    ord(0) = 1, ord(1) = 6, ord(2) = 3, ord(3) = 2, ord(4) = 3, ord(5) = 6.
  - In the group G = $<Z_{10}^*, \times>$, the orders of the elements are:
    ord(1) = 1, ord(3) = 4, ord(7) = 4, ord(9) = 2.

Ring

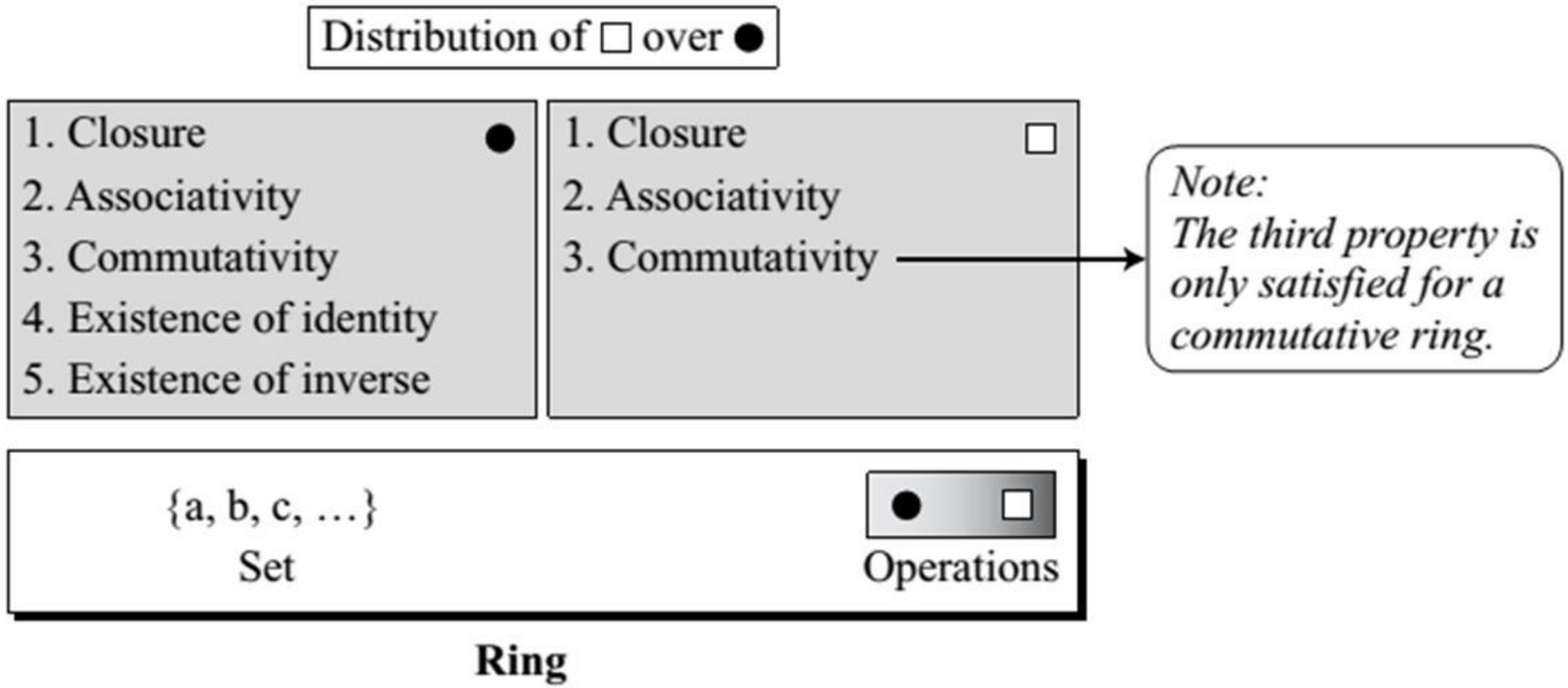
- A ring, R = <{…}, •, ■ >, is an algebraic structure with two operations.
- First operation must satisfy all five properties
- Second operation must satisfy only the first two
- In addition, second operation must be distributed over first

i.e. for all a, b, and c elements of R, we have,

a ■ (b • c) = (a ■ b) • (a ■ c) and

(a • b) ■ c = (a ■ c) • (a ■ c)

Ring


Distribution of □ over ●
1. Closure ●
2. Associativity
3. Commutativity
4. Existence of identity
5. Existence of inverse
1. Closure □
2. Associativity
3. Commutativity
Note:
The third property is only satisfied for a commutative ring.
{a, b, c, …}
Set
● □
Operations
Ring

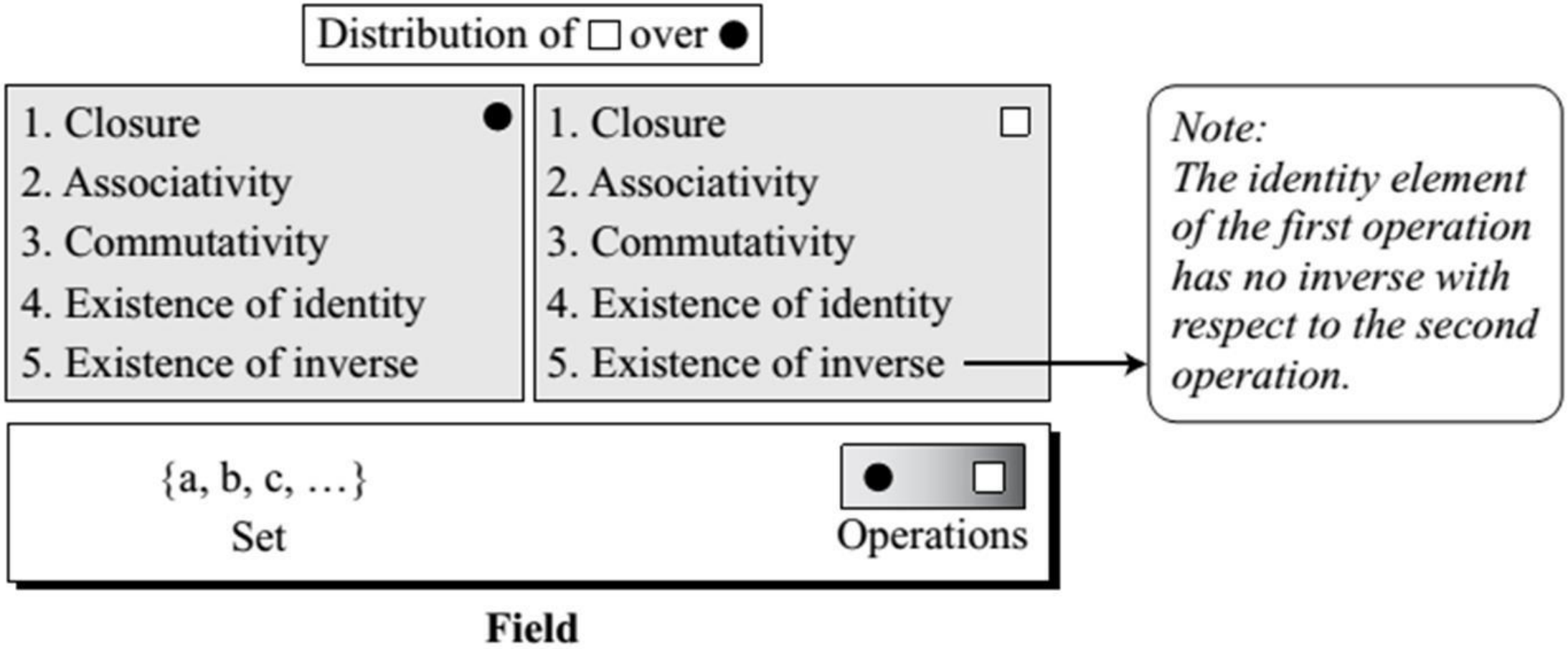
Ring

- The set Z with two operations, addition and multiplication, is a commutative ring.
- We show it by R = <Z, +, ×>.
- Addition satisfies all of the five properties;
- Multiplication satisfies only three properties.
- For example,
  - $5 \times (3 + 2) = (5 \times 3) + (5 \times 2) = 25$.
  - Although, we can perform addition and subtraction on this set, we can perform only multiplication, but not division.
  - Division is not allowed in this structure because it yields an element out of the set.
  - The result of dividing 12 by 5 is 2.4, which is not in the set.

Field

- A field, denoted by F = <{…}, •, ▪ > is a commutative ring in which the second operation satisfies all five properties defined for the first operation except that the identity of the first operation has no inverse.
- Application
  - A field is a structure that supports two pairs of operations that we have used in mathematics: addition/subtraction and multiplication/division.
  - There is one exception: division by zero is not allowed.

Field


Distribution of □ over ●
1. Closure
2. Associativity
3. Commutativity
4. Existence of identity
5. Existence of inverse
●
1. Closure
2. Associativity
3. Commutativity
4. Existence of identity
5. Existence of inverse
□
Note:
The identity element of the first operation has no inverse with respect to the second operation.
{a, b, c, …}
Set
Operations
Field

Fields

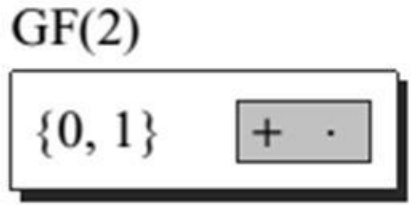
- Finite Fields
  - Galois showed that for a field to be finite, the number of elements should be p^n, where p is a prime and n is a positive integer.

A Galois field, GF(p^n), is a finite field with p^n elements.

- GF(p) Fields
  - When $n = 1$, we have GF(p) field.
  - This field can be the set Z_p, {0, 1, …, p − 1}, with two arithmetic operations.

Fields

- A very common field in this category is GF(2) with the set {0, 1} and two operations, addition and multiplication.



+	0	1
0	0	1
1	1	0

Addition

·	0	1
0	0	0
1	0	1

Multiplication

a	0	1
$-a$	1	0

a	0	1
a^{-1}	—	1

Inverses

Addition/subtraction in GF(2) is the same as the XOR operation; multiplication/division is the same as the AND operation.

Fields

- We can define GF(5) on the set Z_5 (5 is a prime) with addition and multiplication operators.

GF(5)

{0, 1, 2, 3, 4} + ×

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Addition

×	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplication

Additive inverse

a	0	1	2	3	4
−a	0	4	3	2	1

a	0	1	2	3	4
a^{-1}	—	1	3	2	4

Multiplicative inverse

GF(5) field

Summary

Algebraic Structure	*Supported Typical Operations*	*Supported Typical Sets of Integers*
Group	$(+ \ -)$ or $(\times \ \div)$	$\mathbf{Z}_n$ or $\mathbf{Z}_n^*$
Ring	$(+ \ -)$ and $(\times)$	$\mathbf{Z}$
Field	$(+ \ -)$ and $(\times \ \div)$	$\mathbf{Z}_p$

GF(2^n) FIELDS

- In cryptography, we often need to use four operations (addition, subtraction, multiplication and division). In other words, we need to use fields.
- However, when we work with computers, the positive integers are stored in the computers as n-bit words in which n is usually 8, 16, 32 and so on.
- Range of integers is 0 to $2^n - 1$. Hence, the modulus is 2^n.
- So we have two choices if we want to use a field!!!!
  - We can use GF(p) with the set Z_p, where p is the largest prime number less than 2^n.
  - We can work in GF(2^n) and uses a set of 2^n elements.

GF(2^n) FIELDS

- GF(p) with the set Z_p, scheme works, it is inefficient because we cannot use the integers from p to $2^n - 1$.
- For example,
  - if n = 4, the largest prime less than 2^4 is 13. This means that we cannot use integers 13, 14, and 15.
  - If n = 8, the largest prime less than 2^8 is 251, so we cannot use 251, 252, 253, 254, and 255.
- GF(2^n), the elements in this set are n-bit words.
- For example,
  - if n = 3, the set is {000, 001, 010, 011, 100, 101, 110, 111}
- 2^n is not prime. So, we need to define a set of n-bit words and two new operations that satisfies the properties defined for a field.

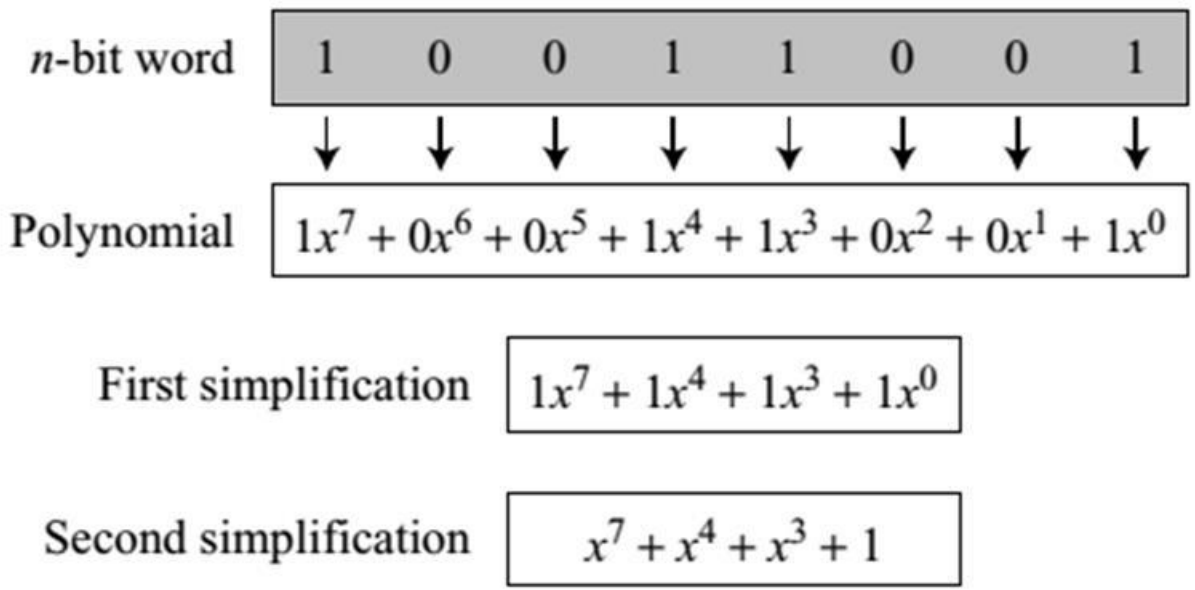
Polynomials

- A polynomial of degree $n - 1$ is an expression of the form

$$f(x) = a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \cdots + a_1x^1 + a_0x^0$$

where x^i is called the ith term and a_i is called coefficient of the ith term.

- We can represent the 8-bit word (10011001) using a polynomial.



Polynomials

- Find the 8-bit word related to the polynomial $x^5 + x^2 + x$, we first supply the omitted terms.
- Since $n = 8$, it means the polynomial is of degree 7. The expanded polynomial is,

$$0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0$$

- This is related to the 8-bit word 00100110.

Polynomials Operations

- Operations on polynomials
  - Actually involves two operations
- Operation on coefficients and operation on polynomials
- Hence, need to define two fields for each
- What for coefficient??
  - Coefficients are made of 0 or 1; we can use the GF(2) field for this purpose.
- What for polynomials???
  - For the polynomials we need the field GF(2^n).

Polynomials

- Modulus
  - For the sets of polynomials in GF(2^n), a group of polynomials of degree *n* is defined as the modulus.
  - Such polynomials are referred to as irreducible polynomials.
- Irreducible polynomials.
  - Prime Polynomial: No polynomial in the set can divide this polynomial
  - Can not be factored into a polynomial with degree of less than n

Degree	*Irreducible Polynomials*
1	$(x + 1)$, (x)
2	$(x^2 + x + 1)$
3	$(x^3 + x^2 + 1)$, $(x^3 + x + 1)$
4	$(x^4 + x^3 + x^2 + x + 1)$, $(x^4 + x^3 + 1)$, $(x^4 + x + 1)$
5	$(x^5 + x^2 + 1)$, $(x^5 + x^3 + x^2 + x + 1)$, $(x^5 + x^4 + x^3 + x + 1)$, $(x^5 + x^4 + x^3 + x^2 + 1)$, $(x^5 + x^4 + x^2 + x + 1)$

Polynomials

- Polynomial addition
  - ***Addition and subtraction operations on polynomials are the same operation***
  - Adding two polynomials of degree n − 1 always create a polynomial with degree n − 1, which means that we do not need to reduce the result using the modulus.
  - In GF(2), A+B=A-B because A-B can be written as A+(-B) and -1=1(mod2) = > -B could be written as B. Therefore, A-B=A+B.
- Example:
  - Let us do $(x^5 + x^2 + x) \oplus (x^3 + x^2 + 1)$ in $GF(2^8)$.
  - We use the symbol $\oplus$ to show that we mean polynomial addition. The following shows the proce

$$
\begin{array}{ll}
0x^7 + 0x^6 + 1x^5 + 0x^4 + 0x^3 + 1x^2 + 1x^1 + 0x^0 & \oplus \\
0x^7 + 0x^6 + 0x^5 + 0x^4 + 1x^3 + 1x^2 + 0x^1 + 1x^0 & \\
\hline
0x^7 + 0x^6 + 1x^5 + 0x^4 + 1x^3 + 0x^2 + 1x^1 + 1x^0 & \rightarrow \quad x^5 + x^3 + x + 1
\end{array}
$$

Polynomials

- Short cut method
  - Addition in GF(2) means the exclusive-or (XOR) operation.
  - So we can exclusive-or the two words, bits by bits, to get the result.
  - In the previous example, $x^5 + x^2 + x$ is 00100110 and $x^3 + x^2 + 1$ is 00001101.
  - The result is 00101011 or in polynomial notation $x^5 + x^3 + x + 1$.

Polynomials

- Multiplication
  - The coefficient multiplication is done in GF(2).
  - The multiplying x^i by x^j results in x^{i+j}.
  - The multiplication may create terms with degree more than $n - 1$, which means the result needs to be reduced using a modulus polynomial.
- For example
  - Find the result of $(x^5 + x^2 + x) \otimes (x^7 + x^4 + x^3 + x^2 + x)$ in GF(2^8) with irreducible polynomial $(x^8 + x^4 + x^3 + x + 1)$.

$$P_1 \otimes P_2 = x^5(x^7 + x^4 + x^3 + x^2 + x) + x^2(x^7 + x^4 + x^3 + x^2 + x) + x(x^7 + x^4 + x^3 + x^2 + x)$$
$$P_1 \otimes P_2 = x^{12} + x^9 + x^8 + x^7 + x^6 + x^9 + x^6 + x^5 + x^4 + x^3 + x^8 + x^5 + x^4 + x^3 + x^2$$
$$P_1 \otimes P_2 = (x^{12} + x^7 + x^2) \bmod (x^8 + x^4 + x^3 + x + 1) = x^5 + x^3 + x^2 + x + 1$$

Polynomials

- To find the final result, divide the polynomial of degree 12 by the polynomial of degree 8 (the modulus) and keep only the remainder.

$$
\begin{array}{r|l}
 & x^4 + 1 \\
\hline
x^8 + x^4 + x^3 + x + 1 & x^{12} + x^7 + x^2 \\
 & x^{12} + x^8 + x^7 + x^5 + x^4 \\
\hline
 & x^8 + x^5 + x^4 + x^2 \\
 & x^8 + x^4 + x^3 + x + 1 \\
\hline
\text{Remainder} & \boxed{x^5 + x^3 + x^2 + x + 1}
\end{array}
$$

Polynomial division with coefficients in GF(2)

Polynomials

- Example:
  - In GF (2^4), find the inverse of ($x^2 + 1$) modulo ($x^4 + x + 1$).
- Solution
  - The answer is ($x^3 + x + 1$)

q	r_1	r_2	r	t_1	t_2	t
$(x^2 + 1)$	$(x^4 + x + 1)$	$(x^2 + 1)$	(x)	(0)	(1)	$(x^2 + 1)$
(x)	$(x^2 + 1)$	(x)	(1)	(1)	$(x^2 + 1)$	$(x^3 + x + 1)$
(x)	(x)	(1)	(0)	$(x^2 + 1)$	$(x^3 + x + 1)$	(0)
	(1)	(0)		$(x^3 + x + 1)$	(0)	

Polynomials

- Example:
  - In GF(2^8), find the inverse of (x^5) modulo ($x^8 + x^4 + x^3 + x + 1$)..
- Solution

q	r_1	r_2	r	t_1	t_2	t
(x^3)	$(x^8 + x^4 + x^3 + x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	(0)	(1)	(x^3)
$(x + 1)$	(x^5)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$
(x)	$(x^4 + x^3 + x + 1)$	$(x^3 + x^2 + 1)$	(1)	(x^3)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$
$(x^3 + x^2 + 1)$	$(x^3 + x^2 + 1)$	(1)	(0)	$(x^4 + x^3 + 1)$	$(x^5 + x^4 + x^3 + x)$	(0)
	(1)	(0)		$(x^5 + x^4 + x^3 + x)$	(0)	

Polynomials

- A better algorithm: Obtain the result by repeatedly multiplying a reduced polynomial by *x*.
- For example, instead of finding the result of ($x^2 \otimes$ P2), the program finds the result of ($x \otimes (x \otimes$ P2)).

- Example:
  - Find the result of multiplying P1 = ($x^5 + x^2 + x$) by P_2 = ($x^7 + x^4 + x^3 + x^2 + x$) in GF($2^8$) with irreducible polynomial ($x^8 + x^4 + x^3 + x + 1$)
- Solution
  - We first find the partial result of multiplying x^0, x^1, x^2, x^3, x^4, and x^5 by P_2.
  - Note that although only three terms are needed, the product of $x^m \otimes$ P2 for *m* from 0 to 5 because each calculation depends on the previous result

Polynomials

Powers	*Operation*	*New Result*	*Reduction*
$x^0 \otimes P_2$		$x^7 + x^4 + x^3 + x^2 + x$	No
$x^1 \otimes P_2$	$\boldsymbol{x} \otimes (x^7 + x^4 + x^3 + x^2 + x)$	$x^5 + x^2 + x + 1$	**Yes**
$x^2 \otimes P_2$	$\boldsymbol{x} \otimes (x^5 + x^2 + x + 1)$	$x^6 + x^3 + x^2 + x$	No
$x^3 \otimes P_2$	$\boldsymbol{x} \otimes (x^6 + x^3 + x^2 + x)$	$x^7 + x^4 + x^3 + x^2$	No
$x^4 \otimes P_2$	$\boldsymbol{x} \otimes (x^7 + x^4 + x^3 + x^2)$	$x^5 + x + 1$	**Yes**
$x^5 \otimes P_2$	$\boldsymbol{x} \otimes (x^5 + x + 1)$	$x^6 + x^2 + x$	No
$\mathbf{P_1 \times P_2} = (x^6 + x^2 + x) + (x^6 + x^3 + x^2 + x) + (x^5 + x^2 + x + 1) = \boldsymbol{x^5 + x^3 + x^2 + x + 1}$			

GF(2^n) FIELDS

- Let us define a GF(2^2) field in which the set has four 2-bit words: {00, 01, 10, 11}.
- We can redefine addition and multiplication for this field in such a way that all properties of these operations are satisfied.

Addition

⊕	00	01	10	11
00	00	01	10	11
01	01	00	11	10
10	10	11	00	01
11	11	10	01	00

Identity: 00

Multiplication

⊗	00	01	10	11
00	00	00	00	00
01	00	01	10	11
10	00	10	11	01
11	00	11	01	10

Identity: 01

An example of a GF(2^2) field

Addition table for $GF(2^3)$

⊕	000 (0)	001 (1)	010 (x)	011 ($x+1$)	100 (x^2)	101 x^2+1	110 (x^2+x)	111 (x^2+x+1)
000 (0)	000 (0)	001 (1)	010 (x)	011 ($x+1$)	100 (x^2)	101 (x^2+1)	110 (x^2+x)	111 (x^2+x+1)
001 (1)	001 (1)	000 (0)	011 ($x+1$)	010 (x^2)	101 (x^2+1)	100 (x^2+x)	111 (x^2+x+1)	110 (x^2+x)
010 (x)	010 (x)	011 ($x+1$)	000 (0)	001 (1)	110 (x^2+x)	111 (x^2+x+1)	100 (x^2+x)	101 (x^2+1)
011 ($x+1$)	011 ($x+1$)	010 (x)	001 (1)	000 (0)	111 (x^2+x+1)	110 (x^2+x)	101 (x^2+1)	100 (x^2)
100 (x^2)	100 (x^2)	101 (x^2+1)	110 (x^2+x)	111 (x^2+x+1)	000 (0)	001 (1)	010 (x)	011 ($x+1$)
101 (x^2+1)	101 (x^2+1)	100 (x^2)	111 (x^2+x+1)	110 (x^2+x)	001 (1)	000 (0)	011 ($x+1$)	010 (x)
110 (x^2+x)	110 (x^2+x)	111 (x^2+x+1)	100 (x^2)	101 (x^2+1)	010 (x)	011 ($x+1$)	000 (0)	001 (1)
111 (x^2+x+1)	111 (x^2+x+1)	110 (x^2+x)	101 (x^2+1)	100 (x^2)	011 ($x+1$)	010 (x)	001 (1)	000 (0)

Multiplication table for $GF(2^3)$ with irreducible polynomial $(x^3 + x^2 + 1)$

$\otimes$	000 **(0)**	001 **(1)**	010 **(x)**	011 **($x+1$)**	100 **(x^2)**	101 **(x^2+1)**	110 **(x^2+x)**	111 **(x^2+x+1)**
000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)	000 (0)
001 (1)	000 (0)	001 (1)	010 (x)	011 ($x+1$)	100 (x^2)	101 (x^2+1)	110 (x^2+x)	111 (x^2+x+1)
010 (x)	000 (0)	010 (x)	100 (x)	110 (x^2+x)	101 (x^2+1)	111 (x^2+x+1)	001 (1)	011 ($x+1$)
011 ($x+1$)	000 (0)	011 ($x+1$)	110 (x^2+x)	101 (x^2+1)	001 (1)	010 (x)	111 (x^2+x+1)	100 (x)
100 (x^2)	000 (0)	100 (x^2)	101 (x^2+1)	001 (1)	111 (x^2+x+1)	011 ($x+1$)	010 (x)	110 (x^2+x)
101 (x^2+1)	000 (0)	101 (x^2+1)	111 (x^2+x+1)	010 (x)	011 ($x+1$)	110 (x^2+x)	100 (x^2)	001 (1)
110 (x^2+x)	000 (0)	110 (x^2+x)	001 (1)	111 (x^2+x+1)	010 (x)	100 (x^2)	011 ($x+1$)	101 (x^2+1)
111 (x^2+x+1)	000 (0)	111 (x^2+x+1)	011 ($x+1$)	100 (x^2)	110 (x^2+x)	001 (1)	101 (x^2+1)	010 (x)